

Total marks – 84

Attempt Questions 1 –7

All questions are of equal value.

Question 1 (12 marks). Start on a SEPARATE page.

- | | Marks |
|--|-------|
| (a) Find the acute angle between the lines $2x + y = 17$ and $3x - y = 3$. | 2 |
| (b) The point $P(17,36)$ divides the line joining $A(2,1)$ and $B(5,8)$ externally in the ratio $m : n$. Find m and n . | 2 |
| (c) Solve for x : $\frac{2x-3}{x-2} \geq 1$ | 2 |
| (d) Differentiate $y = \tan^{-1}\sqrt{3x^2-1}$ | 3 |
| (e) Use the substitution $u = \cos x$ to evaluate $\int_0^{\frac{\pi}{3}} \cos^3 x \sin x \, dx$ | 3 |

Question 2 (12 marks). Start on a SEPARATE page.

(a) Find the term independent of x in the expansion of $\left(2x^3 - \frac{1}{x}\right)^{12}$ 2

(b) Evaluate: $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{2x}$ 2

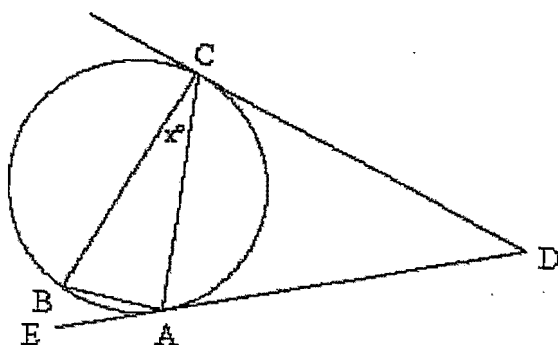
(c) If $f(x) = 2\sin^{-1}3x$, find

(i) the domain and range of $f(x)$. 2

(ii) $f\left(\frac{1}{6}\right)$ 1

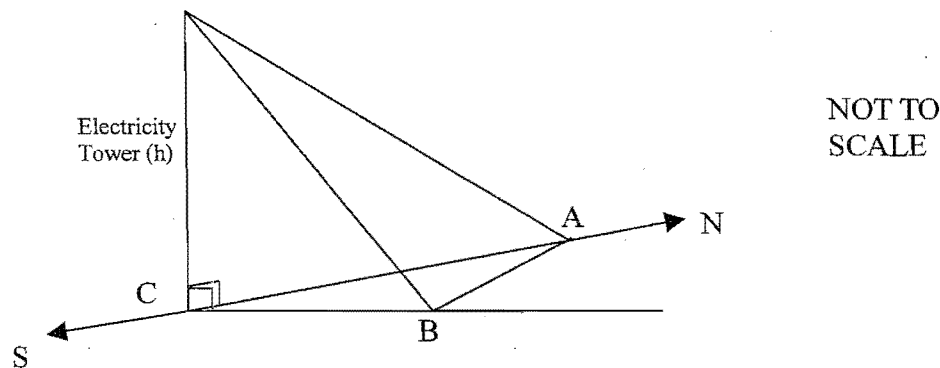
(iii) $f'\left(\frac{1}{6}\right)$ 2

(d) AD and CD are tangents to a circle. B is a point on the circle such that $\angle CBA$ and $\angle CDA$ are equal and are both double $\angle BCA$. Prove that BC is a diameter of the circle. 3



Question 3 (12 marks). Start on a SEPARATE page.

- (a) Leon walks on level ground, in a northerly direction, away from an electricity tower. When he arrives at a point A, the angle of elevation to the top of the tower is 23° . Luke walks on level ground on a bearing of $032^\circ T$ from the same tower, until he reaches point B, and notices that the angle of elevation is 17° . The distance between A and B is 55m. Let h be the height of the tower and assume that the tower base C, is perpendicular to the ground.



- (i) Copy the diagram above onto your booklet and clearly mark on it all the information given. 1
- (ii) Find expressions for AC and BC in terms of h . 2
- (iii) Hence, or otherwise, find the height h of the tower to the nearest metre. 3
- (b) Find how many arrangements can be made by taking all the letters of the word
- (i) MATHEMATICS
- (ii) In how many of them do the vowels occur together? 3
- (c) An archer finds that in the long run, he scores a bull's eye on 3 out of 5 occasions. He fires 8 rounds at a target. Assuming that each trial is an independent event, find the probability of
- (i) exactly 5 bull's eyes.
- (ii) at least 7 bull's eyes. 3

Question 4 (12 marks). Start on a SEPARATE page.

- (a) Prove the following by the Principle of mathematical induction.

$5^{2n} - 1$ is divisible by 24 for $n \geq 1$. 3

- (b) $P(2ap, ap^2)$ is a variable point on the parabola $x^2 = 4ay$. M is the foot of the perpendicular from P to the x -axis. Q is the point on MP such that $MP = PQ$. Find the equation of the locus of Q . 3

- (c) For the function $y = \frac{x^2}{x^2 - 9}$

- (i) Write down the equations of horizontal and vertical asymptotes. 2
- (ii) Find any stationary points and determine their nature. 2
- (iii) Sketch the graph showing the above features. 2

Question 5 (12 marks)

(a) (i) Express $3 \sin x - \sqrt{3} \cos x$ in the form $A \sin(x - \alpha)$ 2

(ii) Hence find the general solution to $3 \sin x - \sqrt{3} \cos x = \sqrt{3}$ 2

(b) N is the number of Kangaroos in a certain population at time t years.

The population size N satisfies the equation

$$\frac{dN}{dt} = -k(N - 500), \text{ for some constant } k.$$

(i) Verify that $N = 500 + Ae^{-kt}$ where A is a constant, is a solution of the equation. 2

(ii) Initially, there are 3500 Kangaroos but after 3 years there are only 3300 left. Find the value of A and the exact value of k . 2

(iii) Find when the number of Kangaroos begin to fall below 2300. 2

(iv) Sketch the graph of the population size against time. 2

Question 6 (12 marks). Start on a SEPARATE page.

- (a) Find the roots of the equation $x^3 - 15x + 4 = 0$, given that two of its roots are reciprocals. 3
- (b) If $\frac{dx}{dt} = x + 6$ and $x = -5$ when $t = 0$, find an expression for x in terms of t . 2
- (c) The speed v m/s of a particle moving in a straight line is given by $v^2 = 64 - 16x - 8x^2$ where the displacement from a fixed point O is x metres.
- (i) Find an expression for the acceleration and show that the motion is simple harmonic. 2
- (ii) Between which two points is the particle oscillating? 2
- (iii) Find the period and amplitude of the motion. 2
- (iv) Find the maximum speed of the particle. 1

Question 7 (12 marks). Start on a SEPARATE page.

- (a) A spherical bubble is expanding so that its volume is increasing at $10 \text{ cm}^3/\text{s}$. Find the rate of increase of its radius when the surface area is 500 cm^2 .

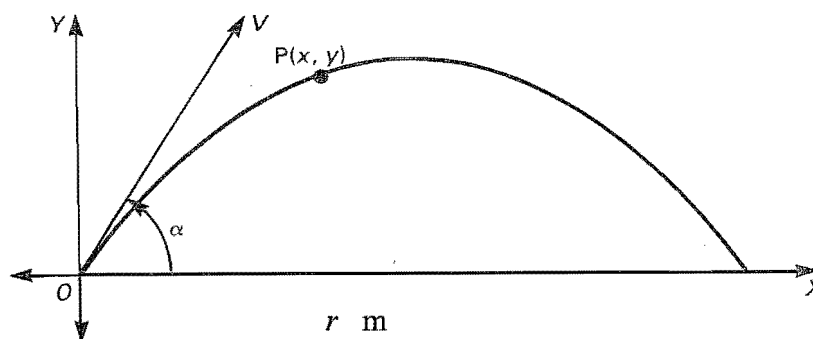
3

- (b) Given that the function $f(x) = \cos x - \log_e x$ has a root between 1.3 and 1.4. Using halving the interval method, find a better approximation to the root correct to one decimal place.

2

- (c) A projectile is fired with initial speed $V \text{ m/s}$ to strike a target on the level ground which is at a distance of $r \text{ m}$ from the origin. The position of the particle at any time t is given by

$$x = Vt \cos \alpha \quad \text{and} \quad y = Vt \sin \alpha - \frac{1}{2}gt^2 \quad (\text{do not prove this})$$



- (i) If α is a suitable angle of projection, prove that

$$\tan^2 \alpha - \left(\frac{2v^2}{gr} \right) \tan \alpha + 1 = 0$$

3

- (ii) Prove that there are two angles of projection if $r < \frac{v^2}{g}$

2

- (iii) Show that the two angles of projection are complementary.
(Hint: consider the product of the roots of the equation in (i))

2

End of paper

Trial HSC - Extension 1 - 2008 Solutions

Question 1 (12 marks)

(a) $2x + y = 17$

$y = -2x + 17$

$m_1 = -2$

$y = 3x - 3 \therefore m_2 = 3$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-2 - 3}{1 - 6} \right| = 1$ (2)

$\theta = 45^\circ$

(b) A(2,1) B(5,8)

$\frac{-5m + 2n}{n - m} = 17$

$-5m + 2n = 17n - 17m$

$12m = 15n$

$m = \frac{15}{12}n$

$\frac{m}{n} = \frac{15}{12}$ (2)

$\frac{m}{n} = \frac{5}{4}$

$m : n = 5 : 4$

(c) $\frac{2x-3}{x-2} \geq 1$

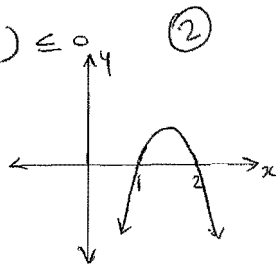
$\frac{(x-2)^2 \times (2x-3)}{x-2} \geq (x-2)^2$

$(x-2)(2x-3) \geq (x-2)^2$

$(x-2)(x-2-2x+3) \leq 0$ (2)

$(x-2)(1-x) \leq 0$

From the graph we get



$x \leq 1$ or $x \geq 2$

(d) $y = \tan^{-1} \sqrt{3x^2 - 1}$

$y' = \frac{1}{1 + 3x^2 - 1} \times \frac{1}{2\sqrt{3x^2 - 1}} \times 6x$ (3)

$= \frac{1}{3x^2} \times \frac{3x}{\sqrt{3x^2 - 1}} = \frac{1}{x\sqrt{3x^2 - 1}}$

(e) $\int_0^{\pi/3} \cos^3 x \sin x \, dx$

Let $u = \cos x$; $\frac{du}{dx} = -\sin x$

$\sin x \, dx = -du$

When $x = 0$, $u = \cos 0 = 1$

When $x = \frac{\pi}{3}$, $u = \cos \frac{\pi}{3} = \frac{1}{2}$

$\int_1^{\frac{1}{2}} u^3 (-du) = -\int_1^{\frac{1}{2}} u^3 \, du$

$= \int_{\frac{1}{2}}^1 u^3 \, du = \frac{u^4}{4} \Big|_{\frac{1}{2}}^1$

$= \frac{1}{4} \left[u^4 \right]_{\frac{1}{2}}^1$ (3)

$= \frac{1}{4} \left(1 - \frac{1}{16} \right) = \frac{15}{64}$

Question 2 (12 marks)

(a)

$T_{r+1} = (-1)^r {}^{12}C_r a^{n-r} b^r$

$T_{r+1} = (-1)^r {}^{12}C_r (2x^3)^{12-r} \left(\frac{1}{x}\right)^r$

$= (-1)^r {}^{12}C_r 2^{12-r} x^{36-3r} \frac{1}{x^r}$

$= (-1)^r {}^{12}C_r 2^{12-r} x^{36-4r}$

For term independent of x

$36 - 4r = 0$

$r = 9$

$T_{10} = (-1)^9 {}^{12}C_9 2^3$ (2)

$= -12 C_9 2^3$

$= -1760$

(b) $\lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{2x}$

$= \lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{\frac{2x}{3}} \times \frac{\frac{2x}{3}}{2x}$ (2)

$= \frac{1}{6} \left(\because \lim_{x \rightarrow 0} \frac{\sin \frac{2x}{3}}{\frac{2x}{3}} = 1 \right)$

(c) $f(x) = 2 \sin^{-1} 3x$

(i) D: $-1 \leq 3x \leq 1$

$-\frac{1}{3} \leq x \leq \frac{1}{3}$ (2)

R: $2x - \frac{\pi}{2} \leq y \leq 2x + \frac{\pi}{2}$

$-\pi \leq y \leq \pi$

(ii) $f\left(\frac{1}{6}\right) = 2 \sin^{-1} \left(3 \times \frac{1}{6}\right)$

$= 2 \sin^{-1} \frac{1}{2}$ (1)

$= 2 \times \frac{\pi}{6} = \frac{\pi}{3}$

(iii) $f'(x) = 2 \times \frac{1}{\sqrt{1-9x^2}} \times 3$

$= \frac{6}{\sqrt{1-9x^2}}$

$f'\left(\frac{1}{6}\right) = \frac{6}{\sqrt{1-9 \times \frac{1}{36}}}$

$= \frac{6}{\sqrt{1-\frac{1}{4}}} = \frac{6}{\frac{\sqrt{3}}{2}}$

$= 4\sqrt{3}$ (2)

(d) $\angle CBA = \angle CDA = 2x$ (given)

$\angle DAC = \angle CBA = 2x$ (angle between tangent and chord is equal to angle in the alternate segment)

$\angle DCA = \angle CBA = 2x$ (" " ")

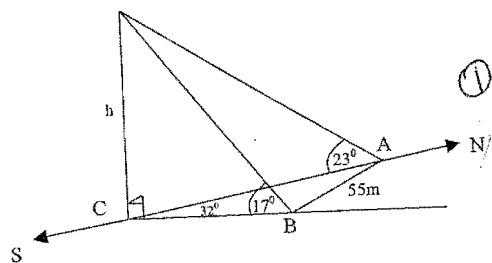
In $\triangle CDA$, $2x + 2x + 2x = 180$ (angle sum of triangle)

$x = 30^\circ$

$\angle BAC = 180 - 3x$
 $= 90$

$\therefore \underline{BC \text{ is diameter}}$ (If the angle subtended at the circumference of a circle by a chord is equal to 90° , then that chord is a diameter.)

Question 3 (12 marks)



(ii) $\tan 67^\circ = \frac{AC}{h}$

$AC = h \tan 67^\circ$

$\tan 73^\circ = \frac{BC}{h}$

$BC = h \tan 73^\circ$ (2)

$55^2 = h^2 \tan^2 67^\circ + h^2 \tan^2 73^\circ$
 $- 2h \tan 67^\circ h \tan 73^\circ \cos 32^\circ$
 $= h^2 (\tan^2 67^\circ + \tan^2 73^\circ - 2 \tan 67^\circ \tan 73^\circ \cos 32^\circ)$

$h^2 = \frac{55^2}{\tan^2 67^\circ + \tan^2 73^\circ - 2 \tan 67^\circ \tan 73^\circ \cos 32^\circ}$

$h = \underline{31\text{m}}$ (3)

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(b) (i) $\frac{11!}{2! 2! 2!} = \underline{4989600}$ (1)

(ii) $\frac{8!}{2! \times 2!} \times \frac{4!}{2!}$
 $= 10080 \times 12$ (2)
 $= \underline{120960}$

(c) (i)

$8C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^3$
 $= \frac{108864}{390625} = \underline{0.279}$

(ii) $8C_7 \left(\frac{3}{5}\right)^7 \left(\frac{2}{5}\right)^1 +$ (3)
 $8C_9 \left(\frac{3}{5}\right)^8$
 $= \frac{41553}{390625} = \underline{0.106}$

Question 4 (12 marks)

(a) Testing $n=1$

$5^{2 \times 1} - 1 = 25 - 1 = 24$ is divisible by 24

$\therefore$ the result is true for $n=1$

Assume the result is true for $n=k$

ie $5^{2k} - 1$ is divisible by 24
 $5^{2k} - 1 = 24P$ where P is an integer (1)

To prove that the result is true for $n=k+1$

ie to prove that $5^{2(k+1)} - 1$ is divisible by 24

ie $5^{2(k+1)} - 1 = 24Q$ where Q is an integer (2)

Now

$5^{2(k+1)} - 1 = 5^{2k+2} - 1$
 $= 5^2 \times 5^{2k} - 1$
 $= 25(24P + 1) - 1$ (by assumpt (1))

$= 600P + 24$

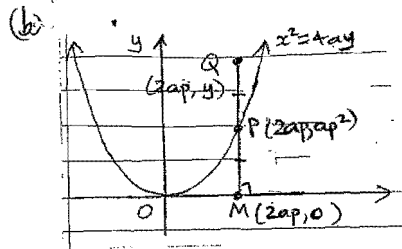
$= 24(25P + 1)$ (3)

$= 24Q$

where $Q = 25P + 1$ is an integer.

Thus the result is true for $n=k+1$

$\therefore$ by the Principle of mathematical induction, the result is true for all $n \geq 1$



Given that $MP = PQ$

$$\sqrt{(ap)^2} = \sqrt{(y - ap^2)^2}$$

$$ap^2 = y - ap^2$$

$$y = 2ap^2$$

The coordinates of Q are

$$x = 2ap \quad \text{--- (1)}$$

$$y = 2ap^2 \quad \text{--- (2)}$$

Squaring (1) we get

$$x^2 = 4a^2 p^2$$

$$= 2a \times 2ap^2$$

$$= 2ay \quad \text{--- (3)}$$

$\therefore$ locus of Q is $x^2 = 2ay$

(c) (i) $\lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 9}$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 - \frac{9}{x^2}} = 1$$

Horizontal asymptote: $y = 1$

Vertical asymptotes are given by $x^2 - 9 = 0$

$$x = \pm 3$$

(2)

(ii) $y = \frac{x^2}{x^2 - 9}$

$$\frac{dy}{dx} = \frac{(x^2 - 9) \times 2x - x^2 \times 2x}{(x^2 - 9)^2}$$

$$= \frac{2x^3 - 18x - 2x^3}{(x^2 - 9)^2}$$

$$= \frac{-18x}{(x^2 - 9)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow -18x = 0 \Rightarrow x = 0$$

when $x = 0$, $y = 0$

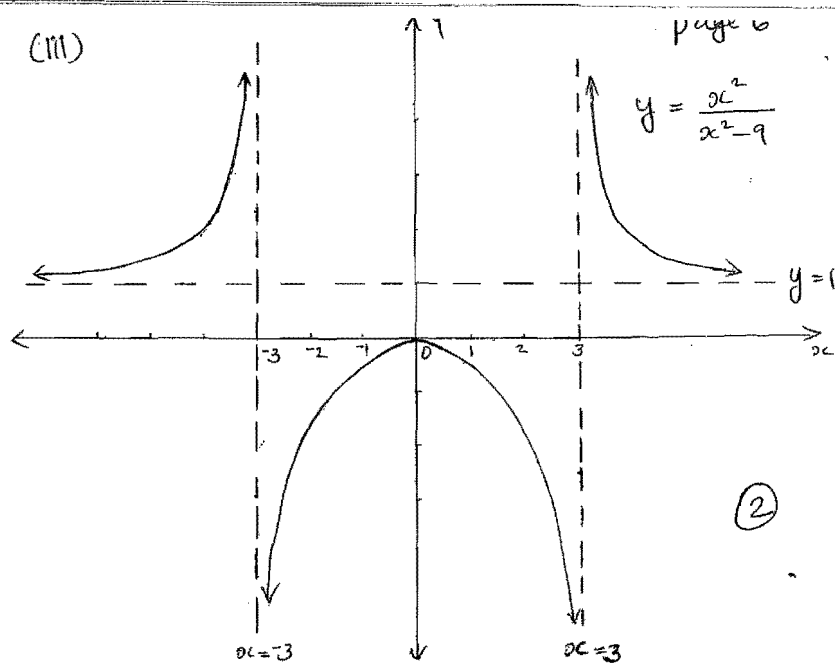
$$\frac{dy}{dx} = \frac{-18x}{(x^2 - 9)^2}$$

when $x = -1$, $\frac{dy}{dx} = \frac{18}{(1 - 9)^2} > 0$

when $x = 1$, $\frac{dy}{dx} = \frac{-18}{(1 - 9)^2} < 0$

$\therefore (0, 0)$ is a maximum turning point. (2)

(iii)



Question 5 (12 marks)

(a) (i)

$$\text{Let } 3 \sin x - \sqrt{3} \cos x = A \sin(x - \alpha)$$

$$= A (\sin x \cos \alpha - \cos x \sin \alpha)$$

$$= A \sin x \cos \alpha - A \cos x \sin \alpha$$

Equating coefficients of $\sin x$ and $\cos x$ we get

$$3 = A \cos \alpha \quad \text{--- (1)}$$

$$\sqrt{3} = A \sin \alpha \quad \text{--- (2)}$$

Squaring and adding (1) and (2) we get

$$A^2 (\sin^2 \alpha + \cos^2 \alpha) = 12$$

$$A^2 = 12 ; A = 2\sqrt{3}$$

Substitute the value of A in (1) and (2)

$$\sin \alpha = \frac{1}{2}$$

$$\cos \alpha = \frac{\sqrt{3}}{2} \quad \therefore \alpha = \frac{\pi}{6}$$

$$\therefore 3 \sin x - \sqrt{3} \cos x = 2\sqrt{3} \sin(x - \frac{\pi}{6}) \quad \text{--- (2)}$$

$$= 2\sqrt{3} \sin(x - \frac{\pi}{6})$$

(ii) $2\sqrt{3} \sin(x - \frac{\pi}{6}) = \sqrt{3}$

$$\sin(x - \frac{\pi}{6}) = \frac{1}{2}$$

General solution is

$$x - \frac{\pi}{6} = n\pi + (-1)^n \frac{\pi}{6}$$

$$x = \frac{\pi}{6} + n\pi + (-1)^n \frac{\pi}{6}$$

(2)

$$b) (i) N = 500 + Ae^{-kt}$$

$$\frac{dN}{dt} = Ae^{-kt} \times -k$$

$$= -kAe^{-kt}$$

$$\text{But } Ae^{-kt} = N - 500 \quad (2)$$

$$\therefore \frac{dN}{dt} = -k(N - 500)$$

$$\therefore N = 500 + Ae^{-kt} \text{ is a solution}$$

$$\text{of } \frac{dN}{dt} = -k(N - 500)$$

$$(ii) \text{ when } t=0, N=3500$$

$$3500 = 500 + A$$

$$A = 3000$$

$$3300 = 500 + 3000e^{-3k}$$

$$e^{-3k} = \frac{2800}{3000} = \frac{14}{15}$$

$$-3k = \log \frac{14}{15} \quad (2)$$

$$k = \frac{-1}{3} \log \frac{14}{15}$$

$$(iii) N < 2300$$

$$500 + 3000e^{-kt} < 2300$$

$$e^{-kt} < \frac{3}{5}$$

$$e^{\frac{1}{3} \log \frac{14}{15} t} < \frac{3}{5}$$

$$\frac{1}{3} \log \frac{14}{15} t < \log \frac{3}{5}$$

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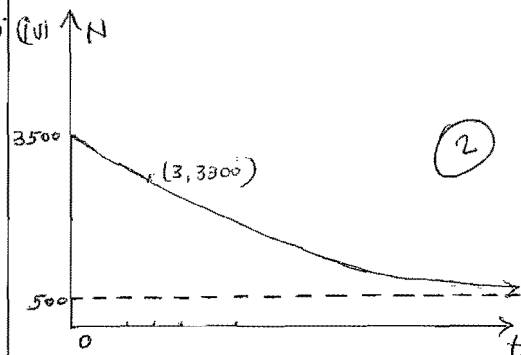
$$t > \log \frac{3}{5}$$

$$\frac{1}{3} \log \frac{14}{15}$$

$$t > 22.21 \text{ years}$$

Number of Kangaroos will begin to fall below 2300 when (2)

$$t > 22.21 \text{ years}$$



Question 6 (12 marks)

(a) Let the roots be $\alpha, \frac{1}{\alpha}$ and β

$$\alpha + \frac{1}{\alpha} + \beta = 0 \quad (1)$$

$$\alpha \times \frac{1}{\alpha} \times \beta = -4 \quad (2) \therefore \beta = -4$$

Substitute in (1)

$$\alpha + \frac{1}{\alpha} - 4 = 0$$

$$\alpha^2 + 1 - 4\alpha = 0$$

$$\alpha^2 - 4\alpha + 1 = 0$$

$$\alpha = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm \sqrt{12}}{2}$$

$$= \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

The roots are

$$2+\sqrt{3}, 2-\sqrt{3} \text{ and } -4 \quad (3)$$

$$(b) \frac{dx}{dt} = x+6$$

$$dt = \frac{1}{x+6} dx$$

$$\int dt = \int \frac{1}{x+6} dx$$

$$t = \log(x+6) + C$$

$$\text{when } t=0, x=-5$$

$$0 = \log(-5+6) + C$$

$$0 = \log(1) + C$$

$$C = 0$$

$$t = \log(x+6) \quad (2)$$

$$e^t = x+6$$

$$x = e^t - 6$$

$$(c) (i) V^2 = 64 - 16x - 8x^2$$

$$\frac{1}{2} V^2 = 32 - 8x - 4x^2$$

$$\frac{d}{dx} \left(\frac{1}{2} V^2 \right) = -8 - 8x$$

$$\ddot{x} = -8(x+1)$$

This is of the form

$$\ddot{x} = -n^2(x-b) \text{ where}$$

$$n = \sqrt{8} \text{ and } b = -1$$

$\therefore$ the motion is (2)

Simple harmonic

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$$(ii) \text{ Let } V^2 = 0$$

$$64 - 16x - 8x^2 = 0$$

$$8x^2 + 16x - 64 = 0$$

$$8(x^2 + 2x - 8) = 0$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0 \quad (2)$$

$$x = -4 \text{ or } 2$$

The particle oscillates between

$$x = -4 \text{ and } x = 2$$

$$(iii) \ddot{x} = -8(x+1)$$

$$n = \sqrt{8}$$

$$\text{Period } T = \frac{2\pi}{n} = \frac{2\pi}{\sqrt{8}}$$

$$= \frac{\pi}{\sqrt{2}} \text{ seconds}$$

$$\text{Amplitude} = 3\text{m} \quad (2)$$

(iv) Maximum speed occurs at the centre

$$\text{i.e. when } x = -1$$

Substitute $x = -1$ in $V^2 = 64 - 16x - 8x^2$ we get

$$V^2 = 64 + 16 - 8$$

$$= 72$$

$$V = \pm \sqrt{72} \quad (1)$$

$$\text{Max. Speed} = \sqrt{72} = 6\sqrt{2} \text{ m/s}$$

Question 7 (12 marks)

$$(a) \frac{dV}{dt} = 10 \text{ cm}^3/\text{s}$$

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$= 4\pi r^2 \frac{dr}{dt}$$

$$10 = 500 \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{10}{500} \quad (3)$$

$$= \frac{1}{50} \text{ cm/s}$$

(b)

$$\begin{array}{ccccccc} + & & - & & - & & - \\ 1.3 & \uparrow & 1.325 & & 1.35 & & 1.4 \\ & 1.3125 & & & & & \end{array}$$

$$\frac{1.3+1.4}{2} = 1.35$$

$$f(1.35) = \cos 1.35 - \log 1.35$$

$$= -0.08$$

The root lies between 1.3 and 1.35

$$\frac{1.3+1.35}{2} = 1.325$$

$$f(1.325) = \cos 1.325 - \log 1.325$$

$$= -0.04$$

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The root lies between 1.3 and 1.325

$$\frac{1.3+1.325}{2} = 1.3125 \quad (2)$$

The approximations to the root are 1.35, 1.325, 1.3125

$\therefore$ the root is 1.3 correct to one decimal place.

(c) (i) when the projectile strikes the target we have

$$Vt \cos \alpha = r \quad (1)$$

$$Vt \sin \alpha - \frac{gt^2}{2} = 0 \quad (2)$$

From (1)

$$t = \frac{r}{V \cos \alpha}$$

Substitute in (2)

$$V \times \frac{r}{V \cos \alpha} \sin \alpha - \frac{g}{2} \times \frac{r^2}{V^2 \cos^2 \alpha} = 0$$

$$r \tan \alpha - \frac{gr^2}{2V^2} \sec^2 \alpha = 0$$

$$r \tan \alpha - \frac{gr^2}{2V^2} (1 + \tan^2 \alpha) = 0$$

$$\frac{gr^2}{2V^2} (1 + \tan^2 \alpha) - r \tan \alpha = 0$$

$$1 + \tan^2 \alpha - \frac{2V^2}{gr^2} r \tan \alpha = 0$$

$$1 + \tan^2 \alpha - \frac{2V^2}{gr} \tan \alpha = 0 \quad (3)$$

$$\tan^2 \alpha - \frac{2V^2}{gr} \tan \alpha + 1 = 0$$

(ii) The quadratic equation

$$\tan^2 \alpha - \frac{2V^2}{gr} \tan \alpha + 1 = 0$$

has real and distinct solutions if $\Delta > 0$

$$\text{i.e. } \left(\frac{2V^2}{gr} \right)^2 - 4 > 0$$

$$\left(\frac{2V^2}{gr} \right)^2 > 4$$

$$\frac{2V^2}{gr} > 2$$

$$2V^2 > 2gr \quad (2)$$

$$V^2 > gr$$

$$\frac{V^2}{g} > r$$

$$\text{i.e. } r < \frac{V^2}{g}$$

(iii) Let $\tan \alpha_1$ and $\tan \alpha_2$ be the roots of the equation

$$\tan^2 \alpha - \frac{2V^2}{gr} \tan \alpha + 1 = 0$$

$$\tan \alpha_1 \times \tan \alpha_2 = 1$$

$$\frac{\sin \alpha_1}{\cos \alpha_1} \times \frac{\sin \alpha_2}{\cos \alpha_2} = 1$$

$$\sin \alpha_1 \sin \alpha_2 = \cos \alpha_1 \cos \alpha_2$$

$$\cos \alpha_1 \cos \alpha_2 - \sin \alpha_1 \sin \alpha_2 = 0$$

$$\cos (\alpha_1 + \alpha_2) = 0$$

$$\alpha_1 + \alpha_2 = 90^\circ \quad (2)$$

$\therefore \alpha_1$ and α_2 are complementary angles.